

MD ENAMUL HOQ

Machine Learning Engineer — Medical Imaging AI · Foundation Models · Clinical Translation & FDA-Ready Validation

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SUMMARY

Machine learning engineer and biomedical-imaging scientist (Ph.D., Biomedical Informatics — Imaging Track) with 5+ years building, training, validating and deploying deep-learning and foundation-model systems for medical imaging across CT and digital pathology. Designs 2D and 3D architectures — CNNs, Vision Transformers, self-supervised foundation models — and data-to-deployment pipelines spanning 50M+ imaging instances, then translates them into clinically interpretable, regulatory-aware tools. Deep experience with DICOM/PACS workflows, multi-institutional and federated datasets, threshold-aware evaluation, and reproducible experimentation aligned with FDA 510(k)/De Novo pathways and Good Machine Learning Practice. Peer-reviewed author at NeurIPS, MIDL and imaging-informatics venues; federated-learning hackathon champion; daily user of LLM coding assistants for well-tested development.

TECHNICAL SKILLS

Machine learning: Model development, training, validation, deployment; supervised, self-supervised and representation learning; classification, detection and quantitative image analysis; imbalanced-data learning; transfer learning and fine-tuning

Architectures: 2D and 3D CNNs; Vision Transformers; self-supervised foundation models (RAD-DINO, DINOv2); diffusion models; multimodal vision-language models; detection frameworks

Imaging & standards: CT; chest-radiograph foundation models; digital pathology (WSI); DICOM, PACS, DICOMweb; volumetric and density quantification; quantitative imaging biomarkers; RadLex, SNOMED; 3D Slicer, ITK-SNAP; TCIA, TCGA, NCI Imaging Data Commons

Languages & frameworks: Python (primary), C/C++, Java, MATLAB; PyTorch, TensorFlow, Keras; NVIDIA FLARE

Evaluation: ROC/AUC, sensitivity/specificity, precision/recall, PR/F1; calibration; threshold and cut-point optimisation; reproducibility, drift and stability analysis

MLOps & practice: Git and collaborative development; code review; testing; experiment tracking and automated reporting; data/model/code versioning and traceability; LLM-based coding assistants

Compute & cloud: Linux; Slurm; HPC with A100 multi-GPU clusters; Google Cloud Platform; AWS; BigQuery

Data engineering: Large-scale imaging pipelines; leakage-safe patient-level splitting; multi-institutional harmonisation and quality control; metadata normalisation; data governance (DUAs, PHI de-identification)

Regulatory & quality: FDA 510(k) and De Novo-oriented documentation; Good Machine Learning Practice; reproducible, auditable validation; clinical-deployment and continuous-monitoring evaluation; IRB-approved prospective validation

EDUCATION

Ph.D., Biomedical Informatics (Imaging Track) — University of Arkansas for Medical Sciences *Expected May 2026*
Little Rock, AR · Dissertation: Foundation Models for Lung Cancer Screening CT (LCS-DINO, native-resolution 2D)

M.S., Engineering Physics — Southeastern Louisiana University *December 2020*
Hammond, LA · Thesis: Image correlation for LDPE film deformation (computational imaging, MATLAB)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (Ph.D. Researcher) — Medical Imaging AI *07/2021 – present*
University of Arkansas for Medical Sciences (UAMS) · Little Rock, AR

- Built an end-to-end lung cancer screening pipeline — DICOM/PACS-compliant curation, RAD-DINO embedding generation, training, patient-level evaluation — over 20M+ CT slices, forming the core of the LCS-DINO foundation-model system.

- Showed that native-resolution 2D foundation-model embeddings are a compute-efficient alternative to full 3D volumetric training for lung cancer classification (first-author study, Journal of Imaging Informatics in Medicine, 2025).
- Designed threshold-aware evaluation — ROC/AUC, PR/F1, prevalence-matched optimal cut-points — reflecting real screening prevalence; reduced false-positive follow-ups in experimental cohorts.
- Scaled multi-GPU training and inference on A100 clusters with Slurm; automated experiment tracking, versioning and report generation for reproducible, auditable results.
- Curated and governed large national cohorts in TCIA and the NCI Imaging Data Commons — cohort definition, manifest generation, DICOMweb/gsutil transfer, BigQuery access, RadLex/SNOMED harmonisation and DUA/PHI governance.
- Collaborated with 15+ radiologists and clinicians to translate model outputs into clinically interpretable metrics and structured reporting; mentored three Ph.D. students and one M.D. student.

Site Lead — Multi-Institutional Federated LDCT Screening (Site 2)

06/2025 – present

UAMS · with Moffitt, Vanderbilt, UCLA and the University of Pittsburgh · Little Rock, AR

- Lead UAMS as Site 2 in a five-institution federated learning programme for low-dose CT lung cancer screening using NVIDIA FLARE.
- Architected privacy-preserving training workflows — secure model updates, cross-site evaluation — that improve detection performance without sharing raw PHI.
- Curated and harmonised multi-site datasets with leakage-safe splits and AI-ready pipelines for cross-institution generalisability and robustness.
- Approach recognised as Champion in three of five categories at the CMU-NVIDIA Federated Learning Hackathon, 2026; work covered by CMU Libraries.

AI & Informatics Research Intern

02/2025 – 07/2025

Mayo Clinic · Rochester, MN

- Developed a multimodal pathology retrieval and auto-captioning framework using vision foundation models and LLMs — top-ten poster, Mayo Clinic AI Summit 2025.
- Raised semantic alignment of generated captions to above 0.82 cosine similarity via a hybrid rule-based and LLM summarisation pipeline, and introduced retrieval-guided generation to reduce unsafe and hallucinated captions.
- Co-led a workshop on fine-tuning and deploying vision-language models in clinical research workflows.

Graduate Research & Teaching Associate; Lab Manager

08/2019 – 12/2020

Southeastern Louisiana University · Hammond, LA

- Developed MATLAB pipelines for digital image correlation with visualisation and modelling tools; supported 13+ researchers with debugging and optimisation; instructed Physics 101/102 laboratories.

Research Intern — Multimedia Lab

09/2018 – 02/2019

National Chung Cheng University · Chiayi, Taiwan

- Developed depth-estimation algorithms for 2D→3D video conversion, improving perceived depth and visual quality.

R&D Engineer

08/2017 – 07/2018

TN Solution · Karlsruhe, Germany & Ho Chi Minh City, Vietnam

- Led image/signal-processing and IoT projects; improved PCB fault diagnosis through noise reduction and feature extraction.

FLAGSHIP PROJECTS — WHAT, HOW, MEASURED IMPACT

LCS-DINO — native-resolution CT foundation model for lung cancer screening (dissertation)

- **What** — End-to-end foundation-model pipeline that classifies lung cancer risk from screening CT at native resolution.
- **How** — Self-supervised RAD-DINO embeddings over 50M+ CT slices; patient-level aggregation with isotonic calibration; prevalence-matched threshold-aware decisioning; full DICOM/PACS curation on A100/Slurm HPC.
- **Impact** — ROC-AUC 0.817 and calibrated PR-AUC 0.705 at ~6% screening prevalence, establishing 2D embeddings as a compute-efficient alternative to 3D volumetric training. First-author publication, J. Imaging Informatics in Medicine, 2025.

Virtual-Eyes — quantitative CT quality control for foundation-model risk prediction

- **What** — Automated QC layer that screens CT quality before foundation-model cancer-risk prediction.

- **How** — Rule-driven quantitative checks on a strict 512×512 16-bit grid, validated against downstream model behaviour on curated screening data.
- **Impact** — Measured, separable reliability gains for generalist versus specialist models, and a preprocessing step that can be written into a protocol. Accepted to MIDL 2026, first author.

Multimodal pathology retrieval & safer auto-captioning (Mayo Clinic)

- **What** — Content-based image retrieval and automated captioning for histopathology whole-slide images.
- **How** — Vision foundation-model embeddings, hybrid rule-based and LLM summarisation, and retrieval-guided generation to ground and constrain outputs.
- **Impact** — Above 0.82 cosine-similarity semantic alignment with fewer unsafe captions; top-ten poster, Mayo AI Summit 2025; published at Ital-IA 2026 (CEUR-WS), first author.

Federated LDCT screening across five institutions (site lead)

- **What** — Privacy-preserving, multi-site model training for low-dose CT lung cancer screening.
- **How** — NVIDIA FLARE; secure model updates; leakage-safe cross-site splits; harmonised multi-institution pipelines.
- **Impact** — Improved detection and reduced projected clinical workload without sharing raw PHI; Champion in three of five categories, CMU-NVIDIA Hackathon 2026.

Synthetic-phantom longitudinal validation of commercial AI (Riverain VS-CT)

- **What** — Framework measuring reliability, measurement reproducibility and drift of a deployed commercial nodule-detection AI over time.
- **How** — MATLAB phantom modelling injecting synthetic pulmonary nodules into real DICOM CT volumes; purpose-built database and analysis workflow; IRB-approved prospective protocol.
- **Impact** — A reproducible, auditable continuous-evaluation methodology for clinical-AI monitoring, aligned with GMLP and post-deployment surveillance expectations.

SELECTED PUBLICATIONS

Full list of 12 works, with links and QR codes, at enamul-hoq.github.io/myWebsite.

- **Hoq ME**, Tarbox L, Johann D Jr, Larson-Prior L, Prior F. Harnessing native-resolution 2D embeddings for lung cancer classification. *J. Imaging Informatics in Medicine*, 2025.
- Alfasy S, Uegami W, **Hoq ME**, Alabtah G, Tizhoosh HR. Semantic and visual crop-guided diffusion models for heterogeneous tissue synthesis in histopathology. *NeurIPS*, 2025.
- **Hoq ME**, Uegami W, Alfasy S, et al. Retrieval-guided generation for safer histopathology image captioning. *Ital-IA 2026, CEUR Workshop Proceedings*.
- **Hoq ME**, Larson-Prior L, Prior F. Virtual-Eyes: quantitative validation of a lung CT quality-control pipeline. *MIDL*, accepted 2026.
- **Hoq ME**, Prior F. Artificial intelligence for low-dose CT lung cancer screening: a systematic review. *Preprints.org*, 2026.

HONORS, SERVICE & CERTIFICATIONS

- Champion, three of five categories — CMU-NVIDIA Federated Learning Hackathon, 2026; top-ten poster, Mayo Clinic AI Summit, 2025; second place, Datathon.org (Emory), 2024; travel grants — SIIM, SCCM, HITLAB.
- Area Chair, Challenges Track — MICCAI 2026. Ethics Reviewer — NeurIPS 2026. Program Reviewer — SIIM Annual Conference, 2025–present. Ad hoc reviewer — Frontiers in Medicine, Journal of Thoracic Disease.
- Contributor to NCI U24 CA215109 (TCIA), two UAMS Team Science Awards (\$100K, \$40K), EU EuCanImage (€805,479) and MOTIE Korea P0006842.
- Certified in Lung Cancer Screening Education (ACR, 2024); National Imaging Informatics Course (SIIM/AAPM/RSNA); 20+ certifications from NVIDIA, Google and Coursera.
- Member — MICCAI Society, SIIM, RSNA, ACR, SCCM.